

EN 502: Emerging Energy Sources

Course Code	EN 502
Course Name	Emerging Energy Sources
L-T-P-C	3-0-0-3
Prerequisites	Instructor's consent
Intended for	UG/PG
Distribution	Compulsory foundation course for M.Tech. (Energy Engineering) and elective for other students
Approval	9th Senate (May 2016)

Course Contents

- **Introduction**

Different types of emerging energy sources, potential and installed capacities of these new generation energy sources, Conversion technologies for different primary energy sources, sustainability benefits, and challenges related with reliability, policy dependence, socio-economic advantages and disadvantages. [6 Lectures]

- **Biomass Energy**

Organic matters available on renewable basis like forests, agricultural, mill and industrial wastes etc., direct fired plants, co fired power plants, gasification, fixed bed gasifiers, small version of gasification or directly fired plants for modular bio power. [8 Lectures]

- **Wind power**

Wind energy availability and basic working principle of wind turbines, wind turbine- rotor blades, tower, nacelle house- electrical generator, power control and other mechanical equipment, resource assessment overview, modern wind turbines, installations and wind farms, advantages and limitations of wind farms . [8 Lectures]

- **Module IV**

Solar power–potential of solar energy reaching earth surface, collecting sunlight, solar photovoltaic and solar thermal techniques, solar cell efficiencies and theoretical limits, solar power plants, future challenges. [8 Lectures]

- **Ocean Energy**

Availability in Indian context, Ocean Thermal Energy Conversion, wave energy conversion, Tidal power basic conversion principle, and challenges related with material corrosion, intermittent primary energy supply, sustainability assessment and improvement. [4 Lectures]

- **Module VI**

Hybrid renewable systems [4 Lectures]

- **Fuel cell**

Proton exchange membrane, PEM chemical reactions, alkaline fuel cells, molten carbonate fuel cell their working reactions and advantages, solid oxide fuel cells- their working reactions and advantages [6 Lectures]

Text Books

1. K. R. Rao, **Energy and Power Generation Handbook: Established and Emerging Technologies**, American Society of Mechanical Engineers, U.S., 2011.
2. Aldo V.da Rosa, **Fundamental of Renewable Energy Processes**, Elsevier Press, 2009.
3. Jahangir Hossain, Mahmud Apel, **Large Scale Renewable Power Generation: Advances in Technologies for Generation, Transmission and Storage (Green Energy and Technology)**, Springer, 2014.
4. Y. Goswami, **Principles of Solar Engineering**, CRC Press, 2013.
5. R. Ehrlich, **Renewable Energy: A First Course**, CRC Press, 2013.
6. D. Spera, **Wind Turbine Technology**, ASME, 2009.
7. S. Srinivasan, **Fuel Cells: From Fundamentals to Applications**, Springer, 2006.
8. D. A. J. Rand and R. M. Dell, **Hydrogen Energy: Challenges and Prospects**, RSC Publishing, 2008.